# Metro Ticket Generating System

## Phase 3-UI/UX Development and Customization

### Interface Design

Interface design focuses on creating a user-friendly and structured layout for the Metro Ticket Booking system. The goal of the interface design phase was to ensure that users can easily navigate the ticket booking process, enter their travel details, and submit ticket requests without confusion.

In this project, the user interface was developed using the ServiceNow Service Catalog, which provides a structured way to design forms and collect user inputs. The interface was designed to be simple, intuitive, and efficient, ensuring that passengers can book tickets quickly.

The interface includes various components such as the Service Catalog item, form variables, and variable types, which together create the ticket booking interface.

**Subtask 1: Creation of Service Catalog**

A Service Catalog Item named Metro Ticket Booking was created in ServiceNow to allow users to request metro tickets through a digital form.

The Service Catalog provides a centralized interface where users can access different services, and in this project it was used to provide a ticket booking interface.

Purpose of the Service Catalog Item

The Metro Ticket Booking catalog item was created to:

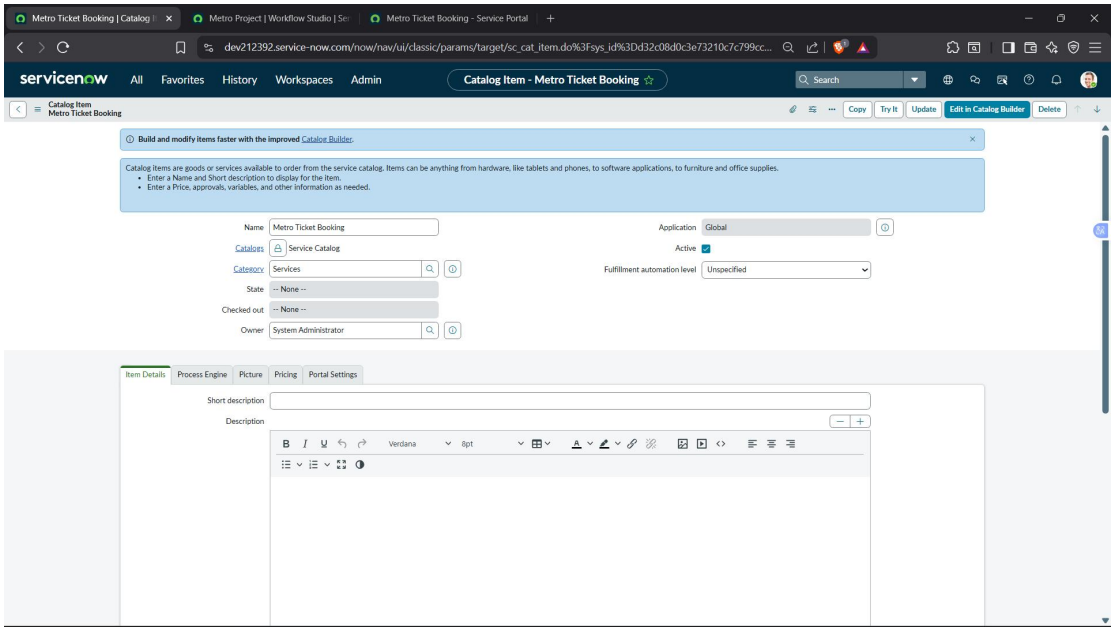
Allow passengers to submit ticket booking requests
Collect travel information such as source station and destination station
Allow users to enter passenger details
Enable automated ticket generation through backend flows

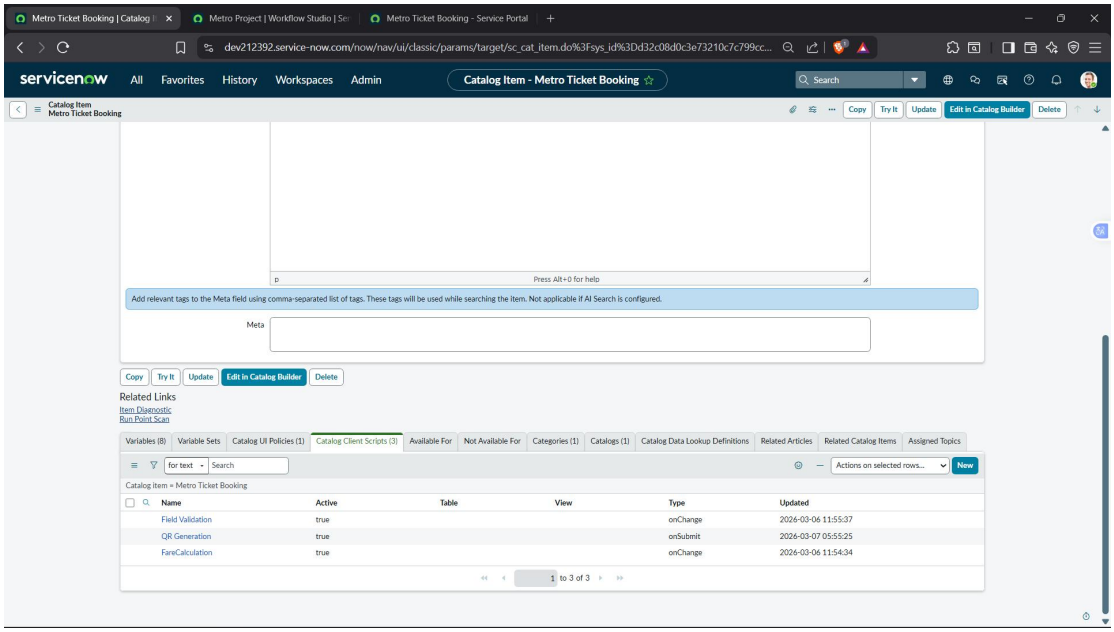
Provide a structured interface for ticket booking

Key Features

Simple ticket booking form
Organized input fields
Integration with backend automation
Easy submission process for users

Once the catalog item is submitted, the system automatically triggers backend automation that creates the metro ticket and generates a QR code.




**Subtask 2: Variables Configuration**

After creating the Service Catalog item, multiple catalog variables were configured to collect user inputs required for booking a metro ticket.

Variables represent the fields that appear in the booking form and allow users to enter their travel information.

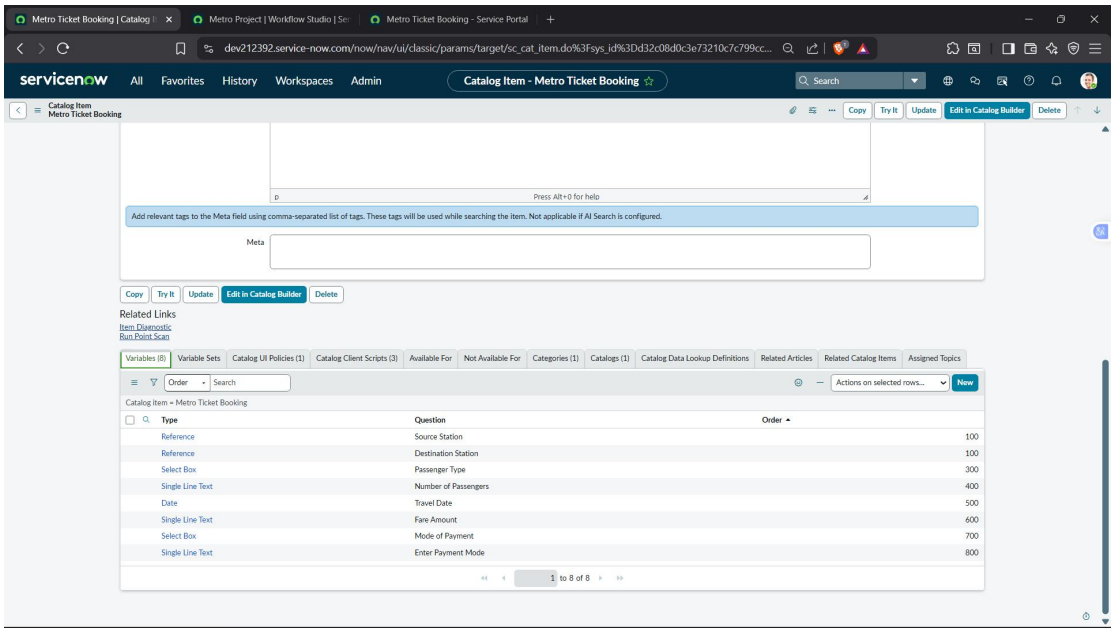
Variables Created

The following variables were configured in the Metro Ticket Booking form:

| Variable Name | Purpose |
|---|---|
| Source Station | Allows the user to select the starting metro station |

| | |
|---|---|
| Destination Station | Allows the user to select the destination station |
| Passenger Type | Specifies whether the passenger is an Adult, Child, or Senior Citizen |
| Number of Passengers | Specifies the number of passengers traveling |
| Travel Date | Allows the user to select the date of travel |
| Fare Amount | Displays the automatically calculated ticket price |
| Mode of Payment | Allows the user to choose the payment method |

These variables help collect all the required information needed to generate a metro ticket.




**Subtask 3: Variable Types Configuration**

Each variable created in the catalog form was assigned an appropriate variable type to ensure proper data input and validation.

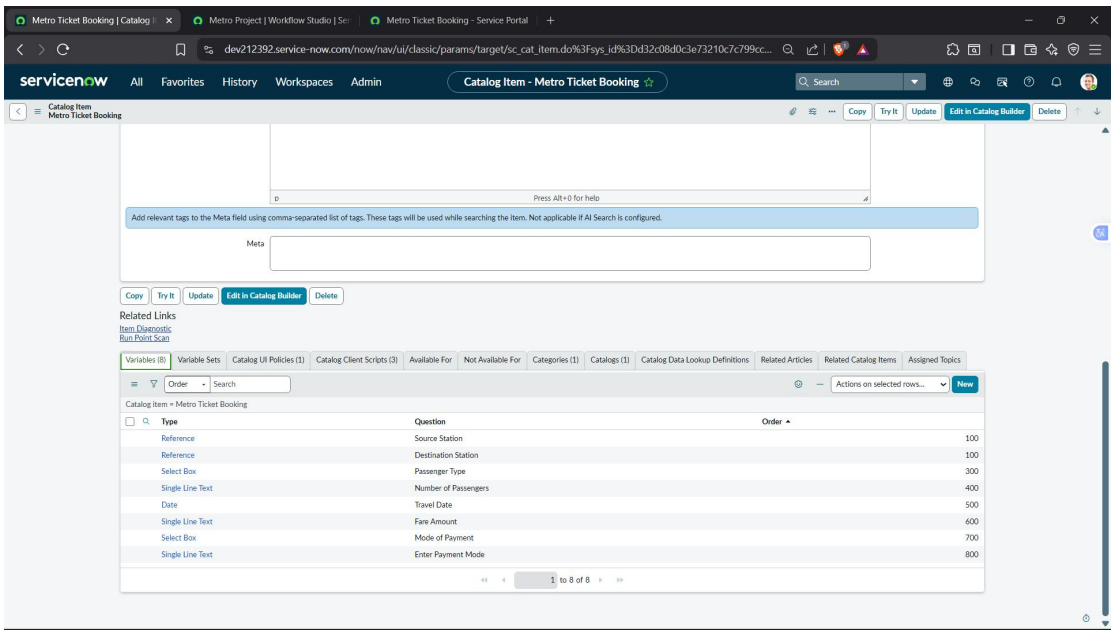
Choosing the correct variable type helps improve user experience and prevents incorrect data entry.

| Variable | Type |
|---|---|
| Source Station | Reference |
| Destination Station | Reference |
| Passenger Type | Choice |
| Number of Passengers | Integer |
| Travel Date | Date |

| | |
|---|---|
| Fare Amount | Integer / Read-only |
| Mode of Payment | Choice |

Purpose of Using Different Variable Types

Reference fields allow users to select stations from a predefined station list.
Choice fields restrict input to predefined options.
Integer fields ensure numeric values are entered.
Date fields allow users to select travel dates using a calendar.
Read-only fields display automatically calculated values like fare.
Using proper variable types ensures data consistency and improves the overall usability of the system.




Summary of Interface Design

The interface design for the Metro Ticket Booking system ensures that users can easily interact with the application and complete the ticket booking process efficiently.

Key UI components implemented include:

Service Catalog Item for ticket booking
User input variables for collecting travel details
Proper variable types for accurate data entry
Integration with backend automation logic

The designed interface provides a simple, structured, and user-friendly experience, making the metro ticket booking process quick and efficient.

## Navigation Flow

Navigation flow describes the sequence of steps that a user follows to access the Metro Ticket Booking system and successfully book a metro ticket. Designing a clear navigation flow is important to ensure that users can easily locate the ticket booking service, enter their travel details, and submit their request without confusion.

In the Metro Ticket Generating System, the navigation flow was designed using the ServiceNow Service Portal, which provides a centralized interface for users to access services through the Service Catalog.

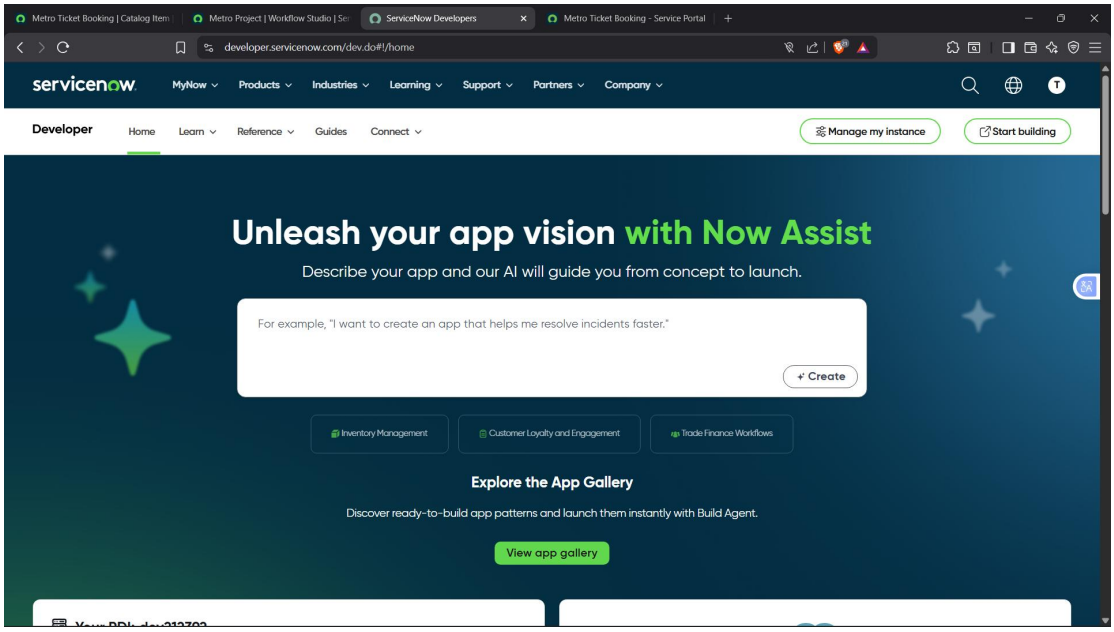
The navigation flow ensures that users can smoothly move from login to ticket booking and ticket confirmation.

**Step 1: Login to ServiceNow Instance**

The first step in the navigation flow is logging into the ServiceNow Personal Developer Instance (PDI) using valid user credentials.

Users must access the ServiceNow instance through the instance URL and authenticate using their username and password.

After successful login, the user is redirected to the ServiceNow homepage where they can access different modules and services.




**Step 2: Access the Service Portal**

After logging in, the user navigates to the Service Portal. The Service Portal provides a user-friendly interface where users can request services and access catalog items.

To access the portal, the user opens the Service Portal URL in the browser.

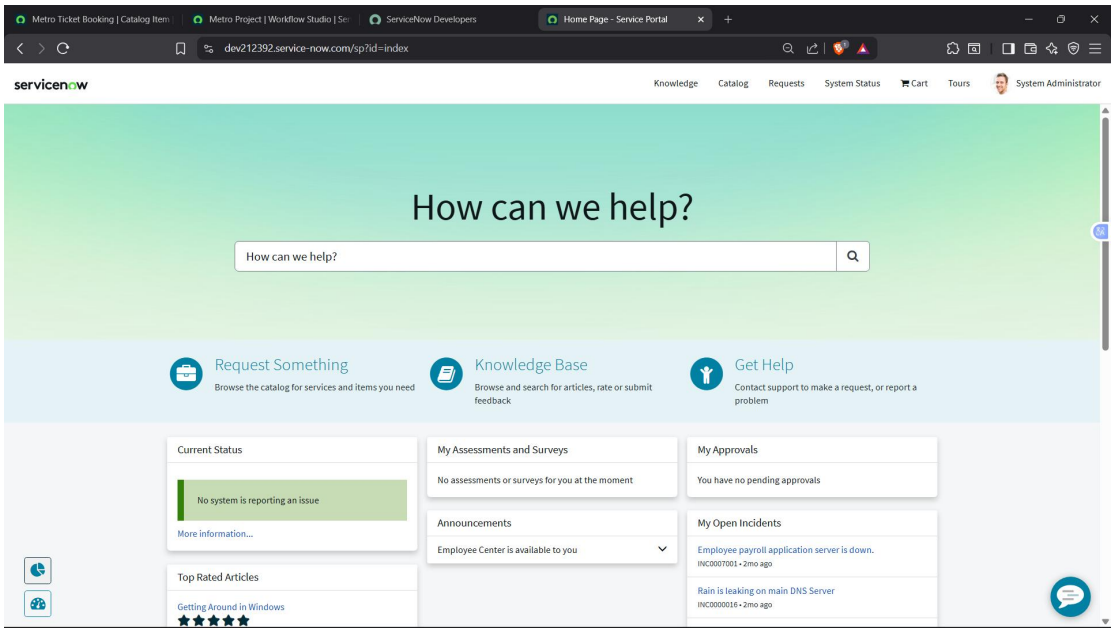
Example format:

https://instance-name.service-now.com/sp

The Service Portal homepage contains features such as:

Service search bar
Catalog categories
Available services
User request tracking

This portal acts as the main interface for the Metro Ticket Booking system.



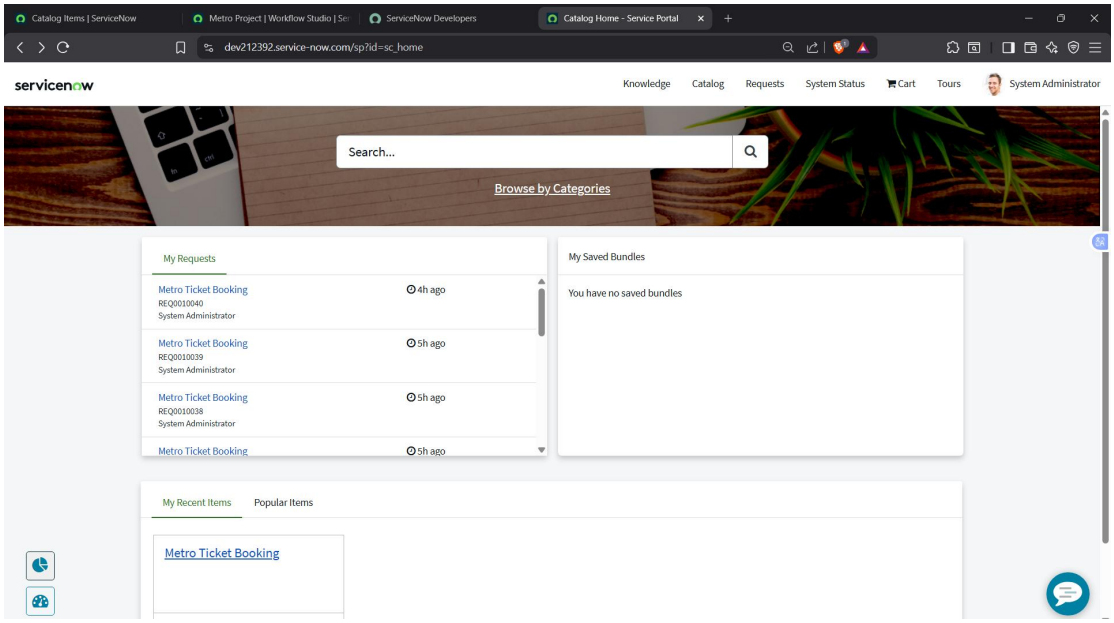

**Step 3: Search for the Metro Ticket Booking Service**

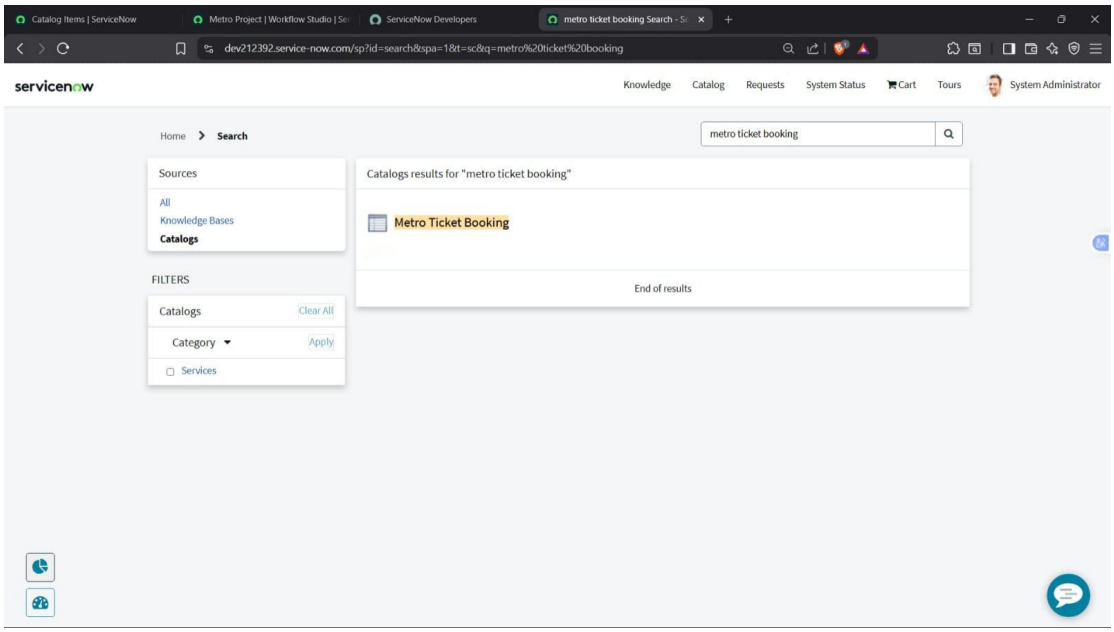
Once inside the Service Portal, the user searches for the Metro Ticket Booking service.

Users can find the service by:

Using the search bar
Navigating through catalog categories
Selecting the service from the available service list

The Metro Ticket Booking catalog item opens the ticket booking form where users can provide travel details.



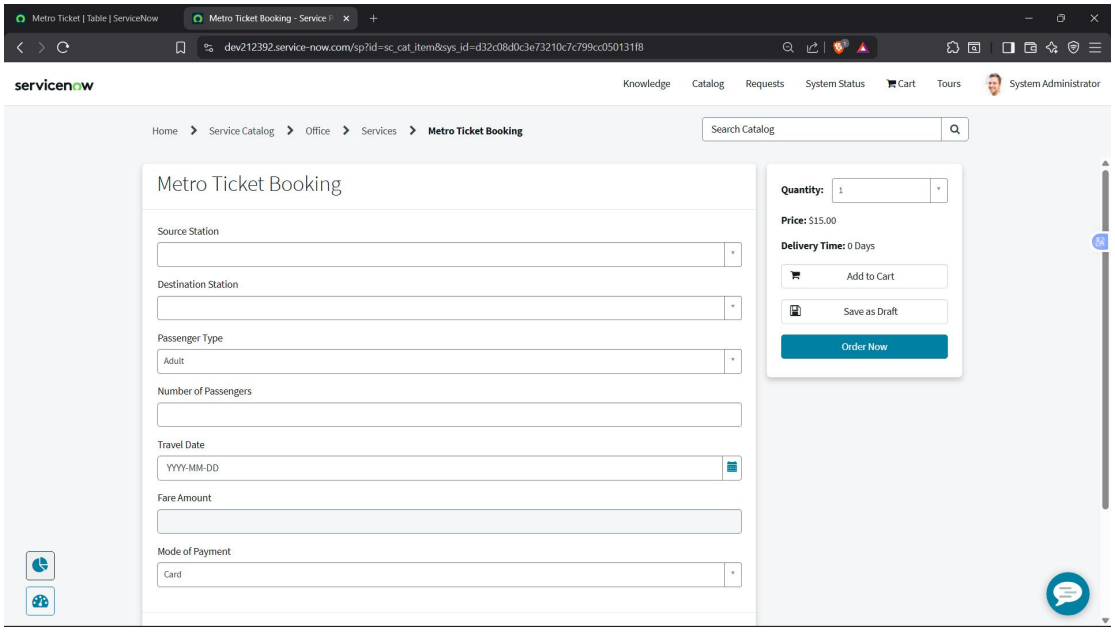

**Step 4: Fill Ticket Booking Details**

After opening the catalog item, the user fills in the required ticket booking details.

The form contains several fields such as:

Source Station
Destination Station
Passenger Type
Number of Passengers
Travel Date
Mode of Payment

As the user fills in the form, automation logic such as fare calculation is triggered dynamically.

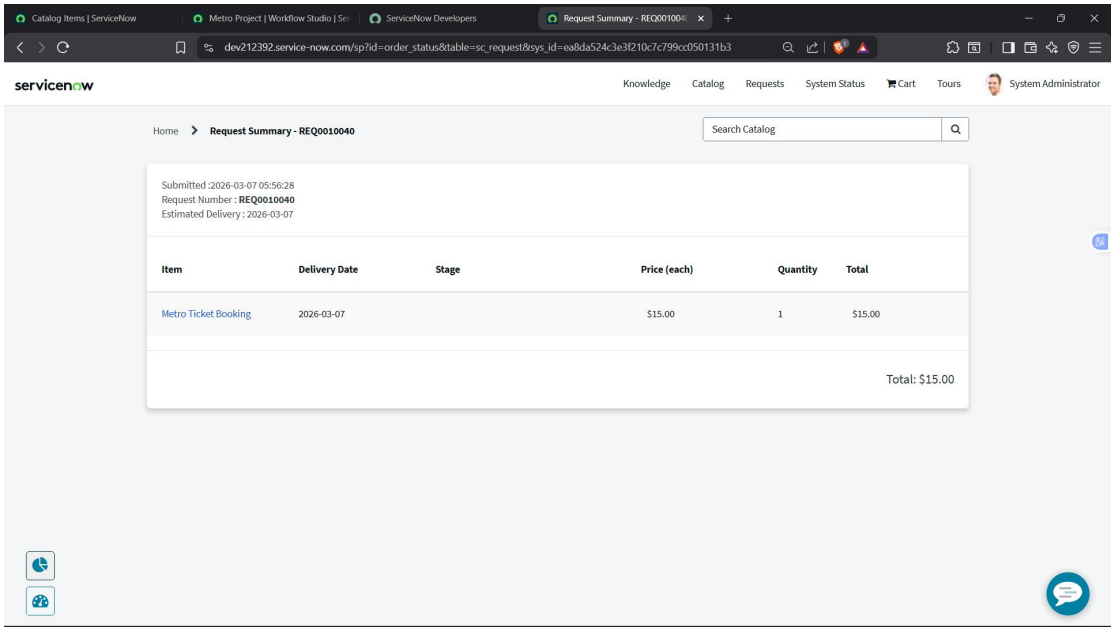



**Step 5: Submit the Ticket Request**

After filling all required details, the user clicks the Order Now / Submit button to complete the ticket booking request.

Once submitted, the system triggers backend automation which performs the following actions:

Creates a new ticket record in the Metro Ticket table
Generates a unique ticket number
Generates a QR code for the ticket
Sends an email confirmation to the user

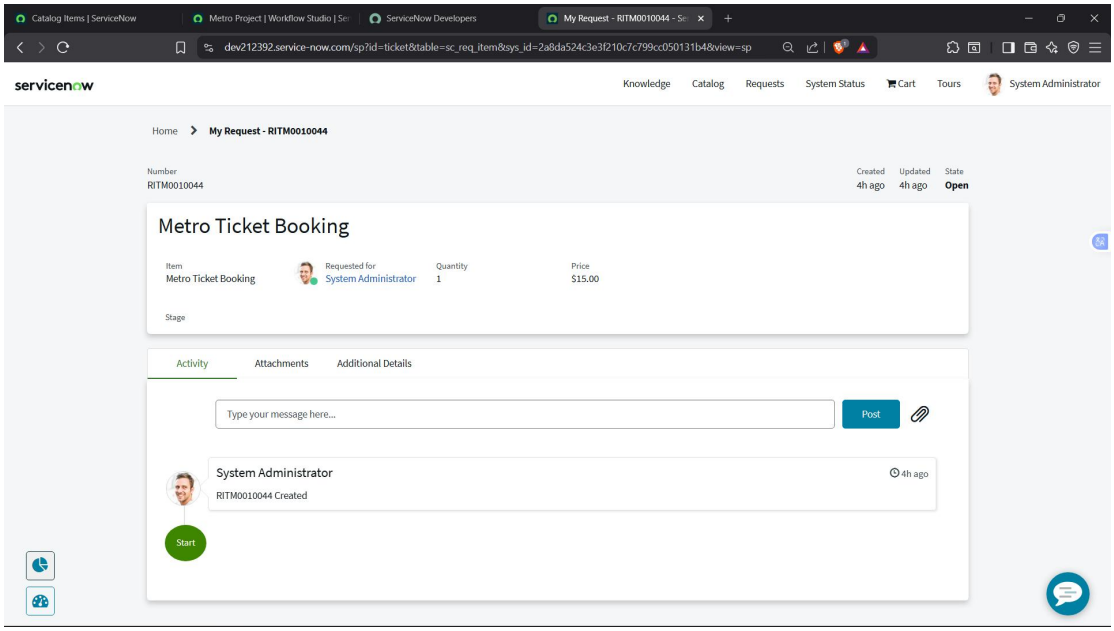


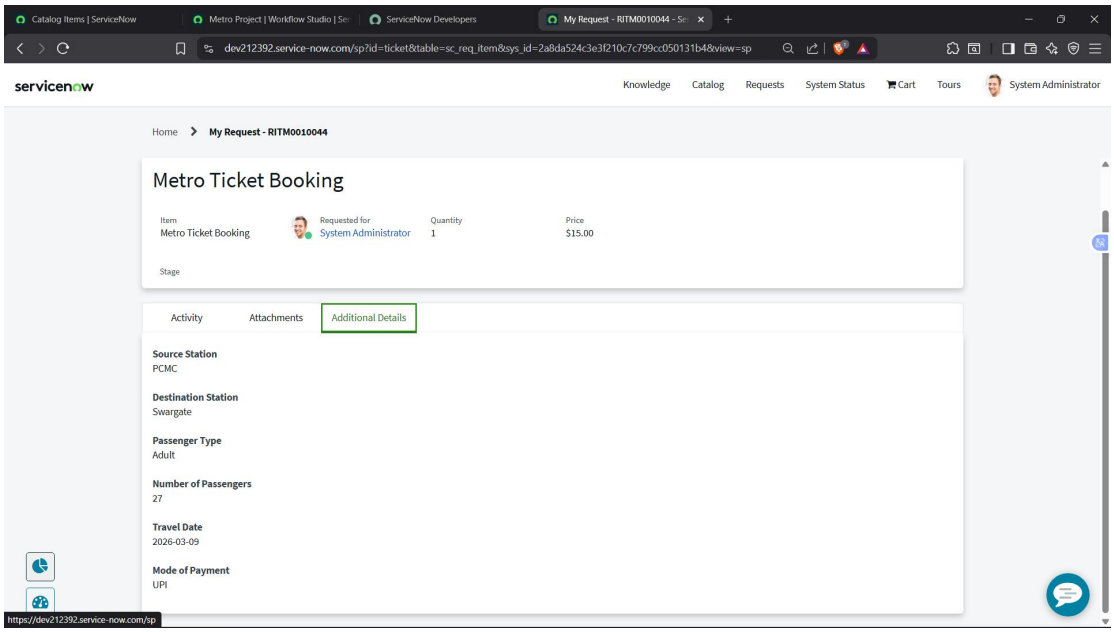

**Step 6: Request Confirmation and Ticket Generation**

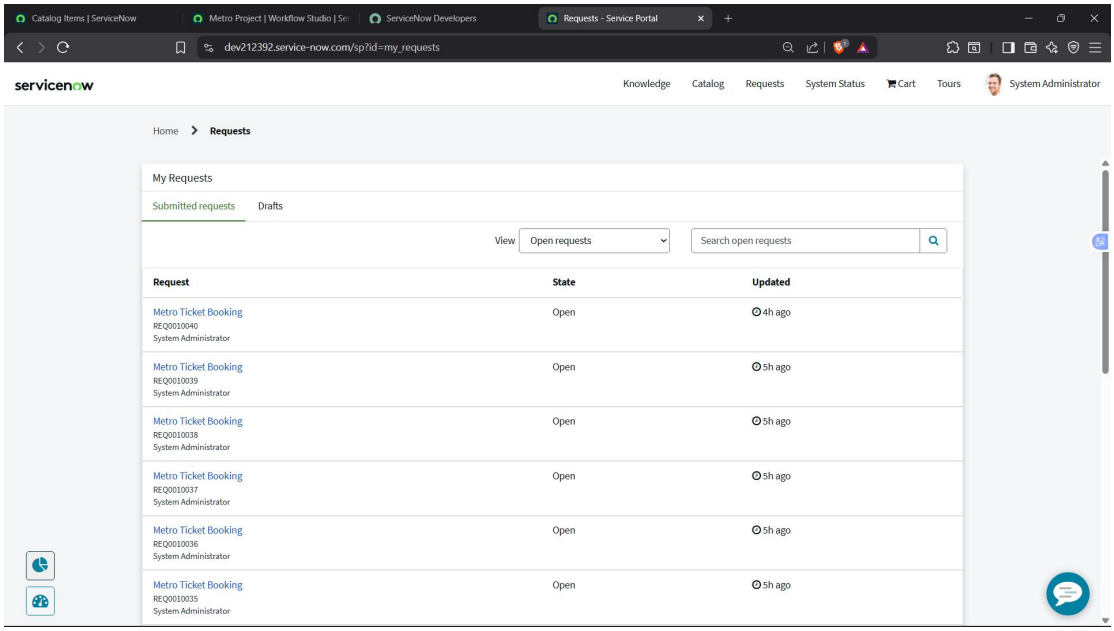
After the request is submitted, the system displays a Request Summary page where the user can see the generated request number and booking details.

The ticket information is stored in the system database and can be accessed through the Requested Item page.

Users can also receive ticket details via email notification.






Navigation Flow Summary

The navigation flow of the Metro Ticket Booking system follows the steps below:

User Login
↓
Access Service Portal
↓
Search Metro Ticket Booking
↓
Open Ticket Booking Form
↓
Fill Travel Details
↓
Submit Request
↓
Ticket Record Generated
↓
QR Code Generated & Email Sent

This structured navigation flow ensures that users can easily book metro tickets through the ServiceNow platform with minimal effort.

## Usability

Usability refers to how easy and efficient it is for users to interact with the system. In the Metro Ticket Generating System, usability was a key focus to ensure that passengers can quickly book tickets without confusion or difficulty.

The interface was designed to provide a simple and intuitive user experience through the ServiceNow Service Portal. The booking form layout, field structure, and automation features were implemented to make the ticket booking process straightforward and user-friendly.

Several usability features were implemented in the system to improve the overall user experience.

Real-Time Fare Preview

One of the major usability features implemented in the system is real-time fare calculation.

When the user selects the Source Station, Destination Station, and Number of Passengers, the system automatically calculates and displays the ticket fare before the form is submitted.

This functionality is implemented using Catalog Client Scripts, which dynamically update the Fare Amount field whenever relevant form values change.

Benefits of Real-Time Fare Preview

Allows users to see the ticket price instantly
Eliminates the need for manual fare calculation
Improves transparency in ticket pricing
Helps users confirm travel costs before submitting the request

This feature enhances the booking experience by providing immediate feedback to users.

User-Friendly Form Layout

The ticket booking form was designed with a clear and organized layout to make it easier for users to understand and fill out the required information.

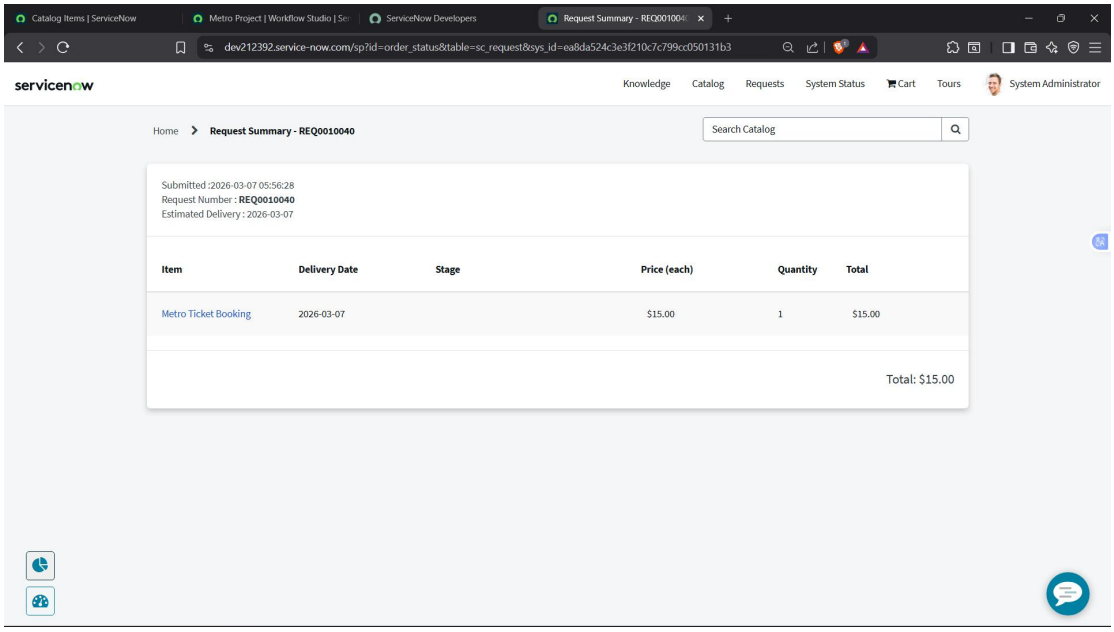
The form includes logically arranged fields such as:

Source Station
Destination Station
Passenger Type
Number of Passengers
Travel Date
Mode of Payment
Fare Amount

These fields are arranged in a structured way so that users can complete the booking process step by step.

Benefits

Reduces user confusion
Improves form readability
Helps users complete the form faster



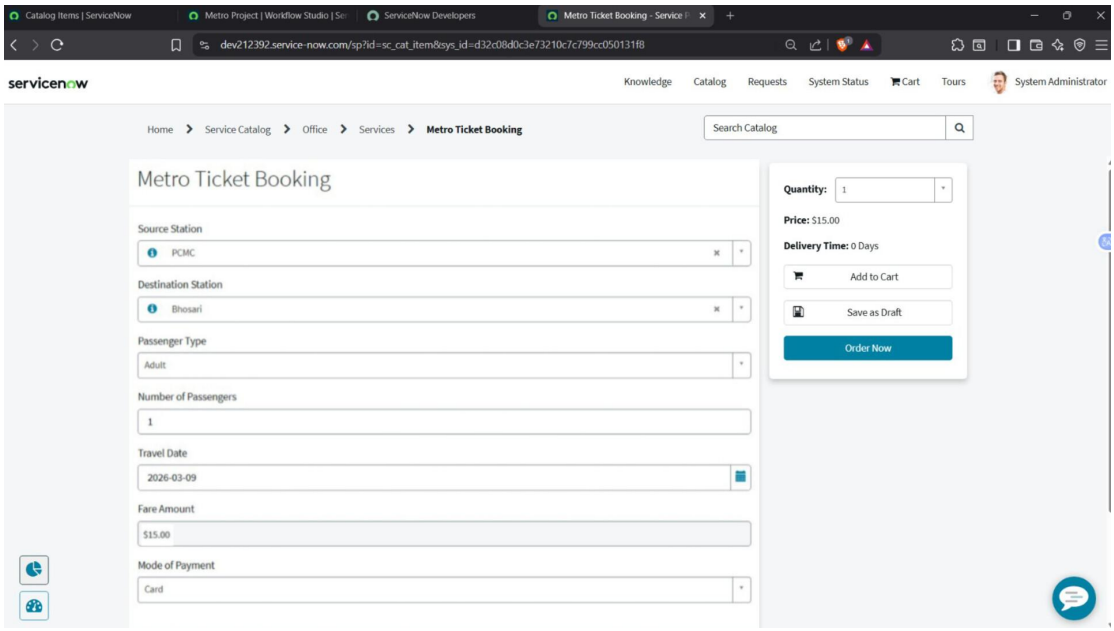

Dynamic Field Interaction

Dynamic form behavior was implemented to make the system more interactive and responsive.
Certain form fields automatically update or respond based on user input. For example:
Fare Amount automatically updates when stations change
Validation prevents incorrect inputs
Certain fields become mandatory before submission

This dynamic behavior ensures that users enter correct information before submitting the ticket request.

Benefits

Prevents incorrect ticket bookings
Improves form accuracy
Enhances system reliability



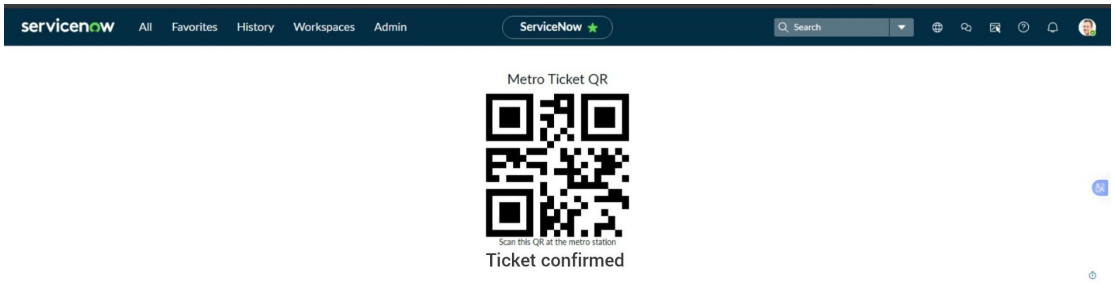
Simplified Ticket Booking Process

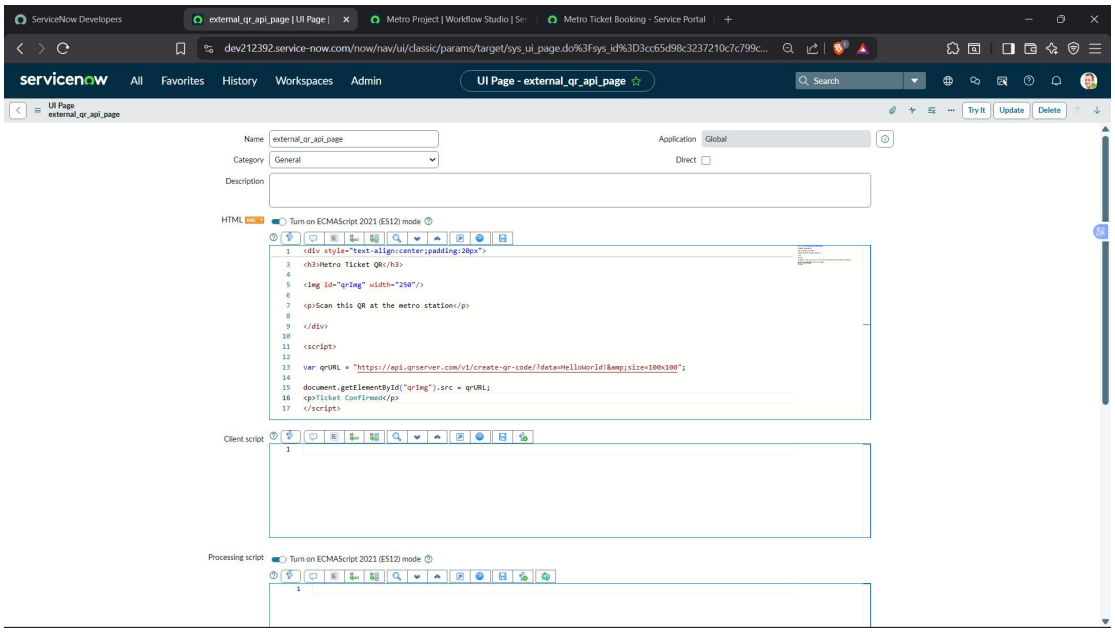
The overall ticket booking process was designed to require minimal steps, allowing users to complete their booking quickly.

The process includes:

Opening the Metro Ticket Booking catalog item
Filling travel details
Viewing calculated fare
Submitting the ticket request
Receiving ticket confirmation and QR code

By minimizing unnecessary steps, the system provides a faster and more efficient ticket booking experience.



Usability Summary

The Metro Ticket Generating System was designed with a strong focus on usability to ensure a smooth experience for passengers. Key usability features implemented include:

Real-time fare calculation
Simple and structured booking form
Dynamic field updates
Clear ticket booking workflow

These improvements help users easily navigate the system and complete ticket bookings efficiently.